Comparative intelligence and ethics



Classical and deep learning
for movie recommendations

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MovieLens 100K

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Inputs: user ID and movie ID

Titles support the demonstration

Historical data collected during 1997–1998

Ratings	Users	Movies
100,000	943	1,682

Whole-star ratings from 1 to 5

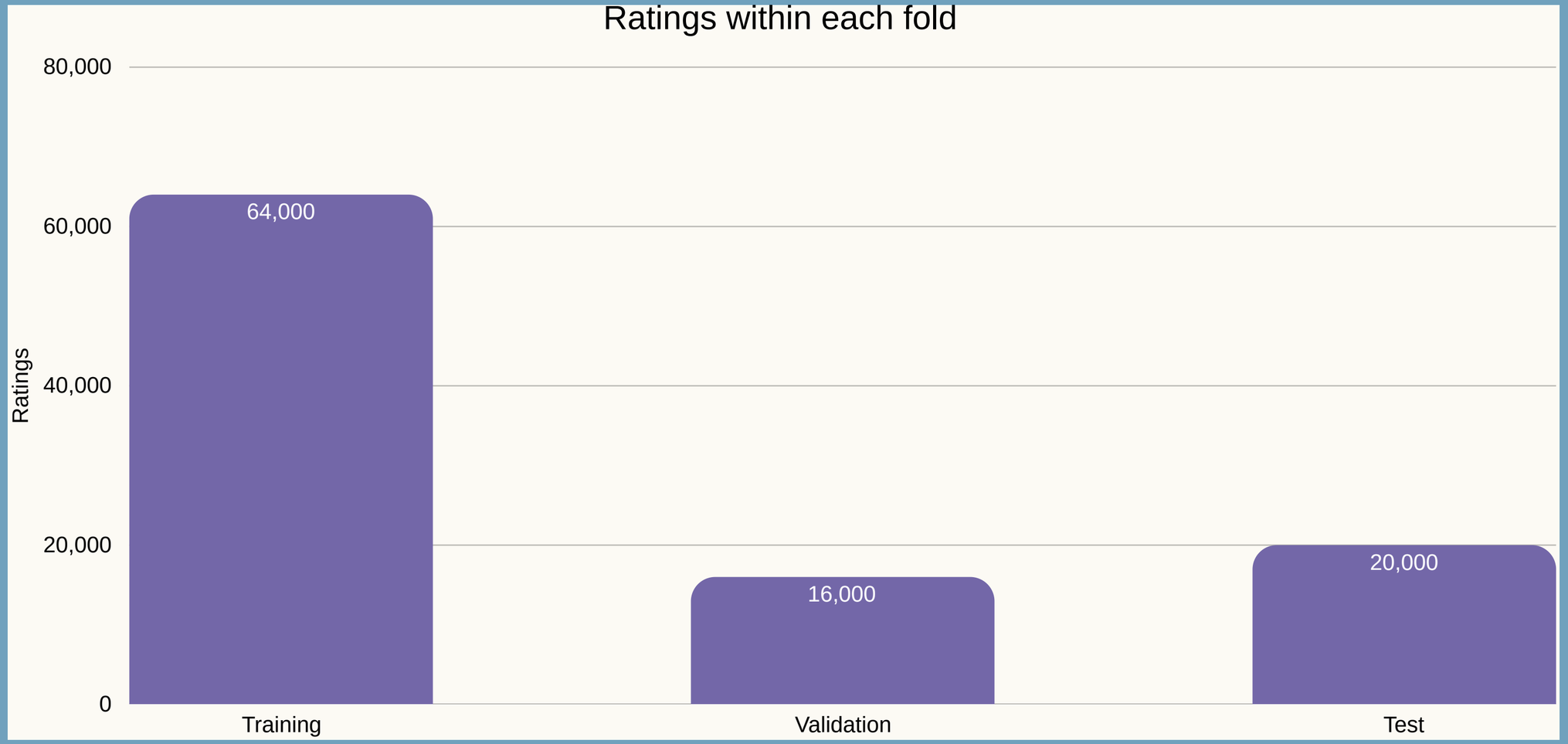
Rating coverage

93.70% of possible pairs have no rating

Rating evidence varies	Users	Movies
Median ratings	65	27
Minimum ratings	20	1
Maximum ratings	737	583

No missing values, duplicate pairs, or ratings outside 1–5.
Unrated means unknown.

Five-fold validation



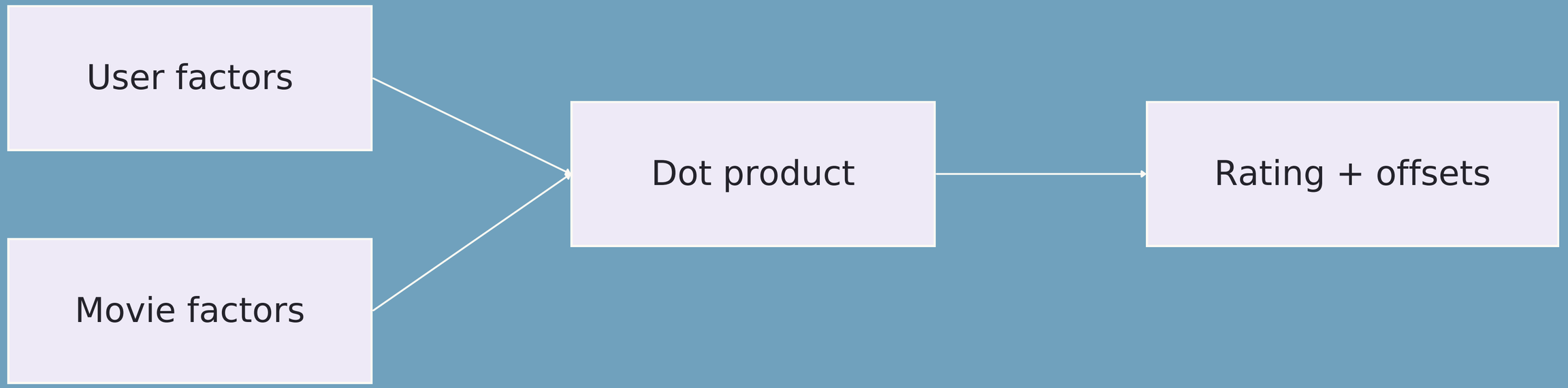
5 folds

Same splits for both models, Users remain in training, Settings chosen per fold

Each rating enters a test subset once. Future behaviour needs chronological evaluation.

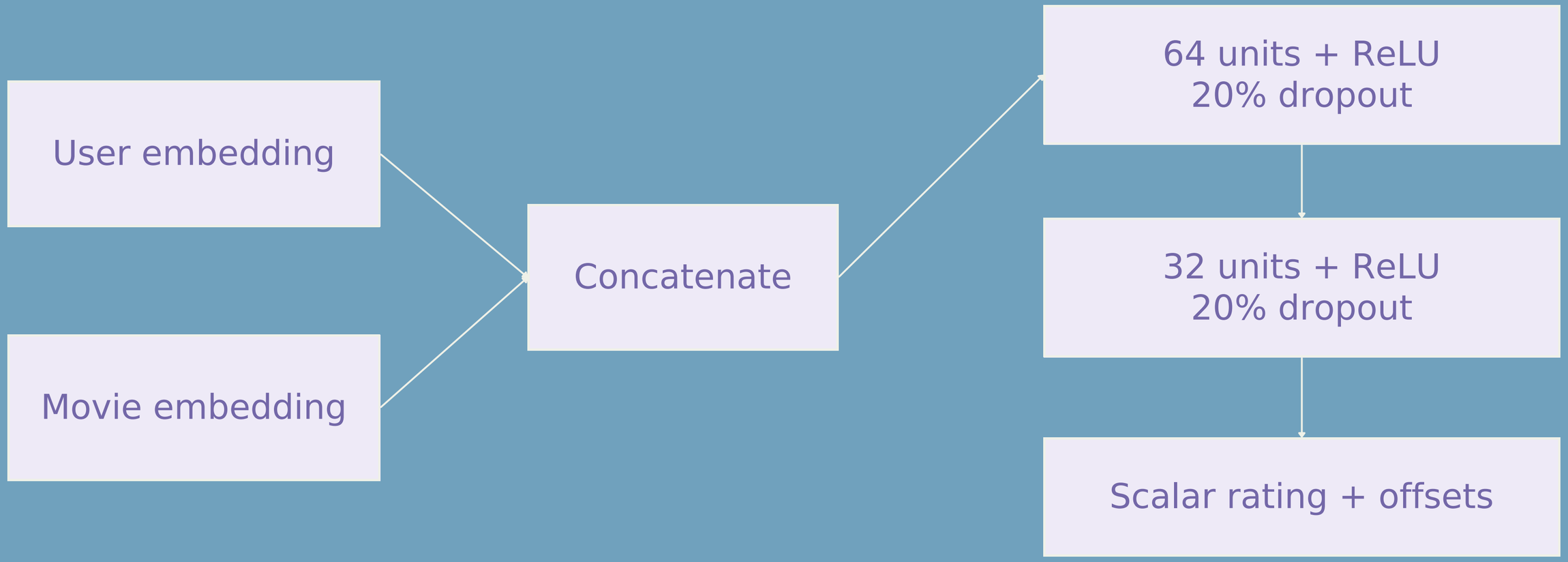
Classical matrix factorisation

Predicted rating = mean + user offset + movie offset + dot product



The dot product models the user–movie interaction. Factors have no predefined genre labels.

Neural recommendation model



The network learns from ratings. Both models use the same identity inputs and rating offsets.

Training and model selection

Setting	Factors	Weight decay
Original	20	0.0001
Stronger L2	20	0.001
Fewer factors	10	0.0001

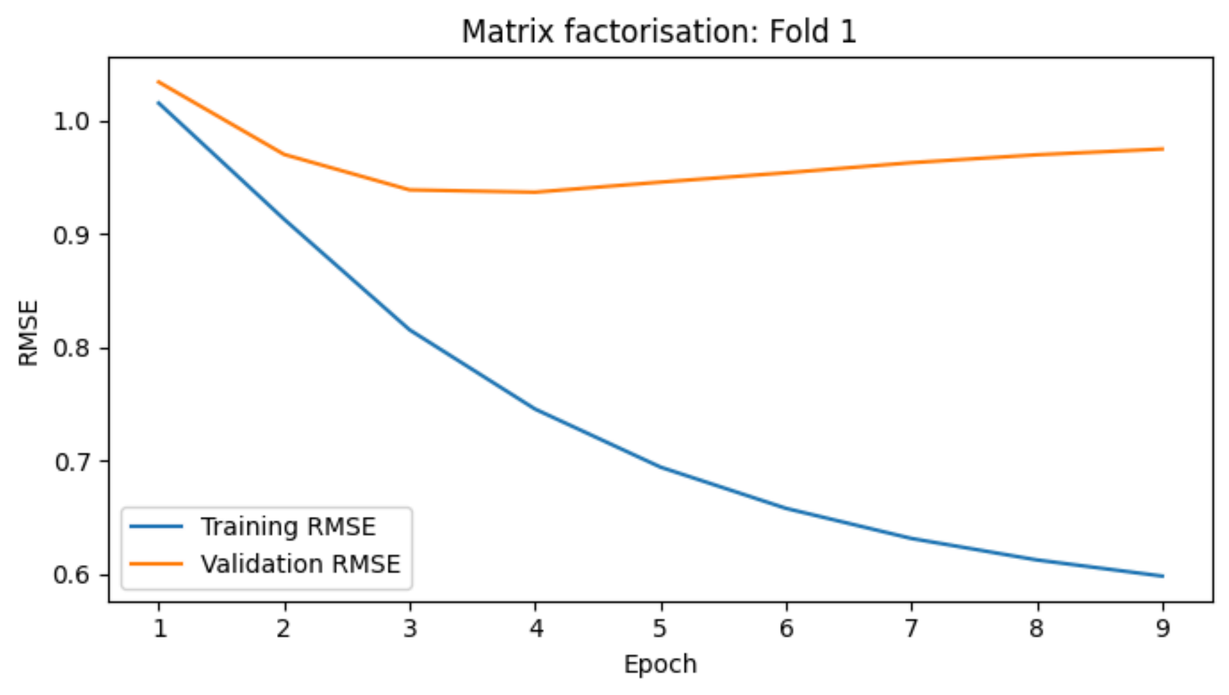
30 training trials

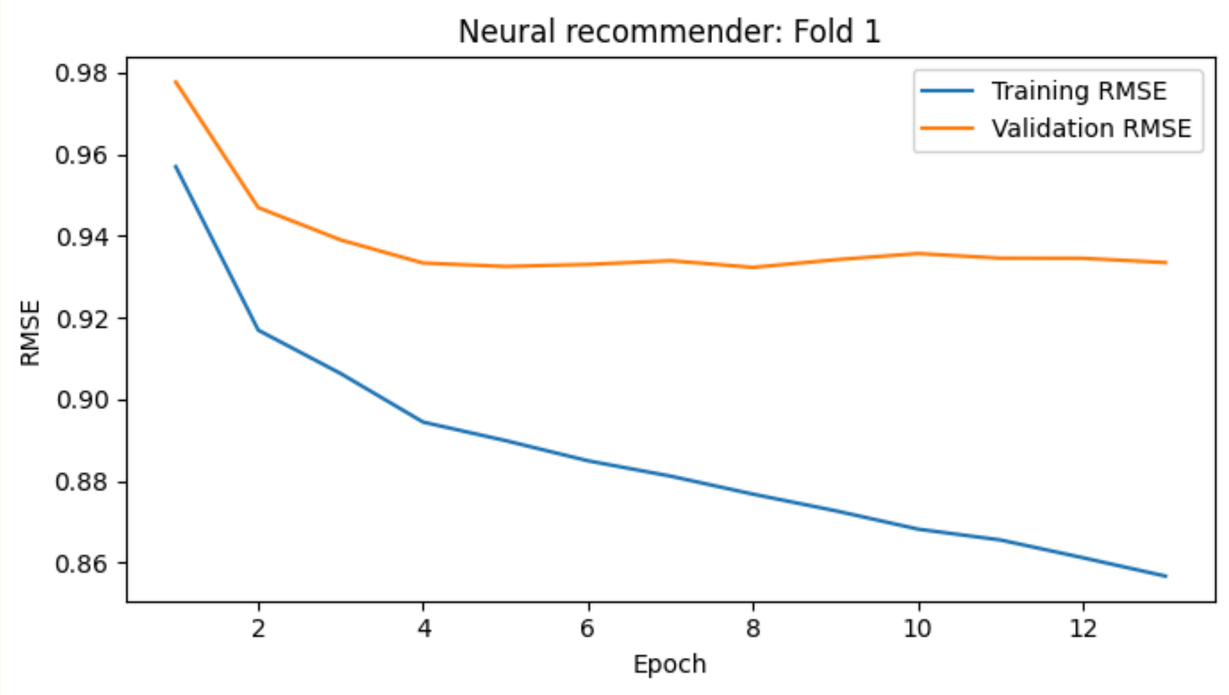
Adam, MSE
Batch size 2,048
Maximum 40 epochs
Patience 5

Learning rate: MF 0.01, neural 0.003
Each fold retains its best validation checkpoint

Three settings per model vary factor count and regularisation.

Overfitting and early stopping





Initial Fold 1 runs. Best checkpoint: MF epoch 4, neural epoch 8.
Each later trial has its own best checkpoint.

Rating accuracy and ranking quality

Measure	Purpose	Direction
RMSE	Rating error with a greater penalty for large mistakes	Lower
MAE	Average absolute rating error	Lower
NDCG@10	Relevant movies near the top	Higher

Ranking candidates: held-out rated movies

Relevant = actual rating 4 or 5
At least 10 candidates and one relevant movie
Eligible users: 61.40–61.93% per fold

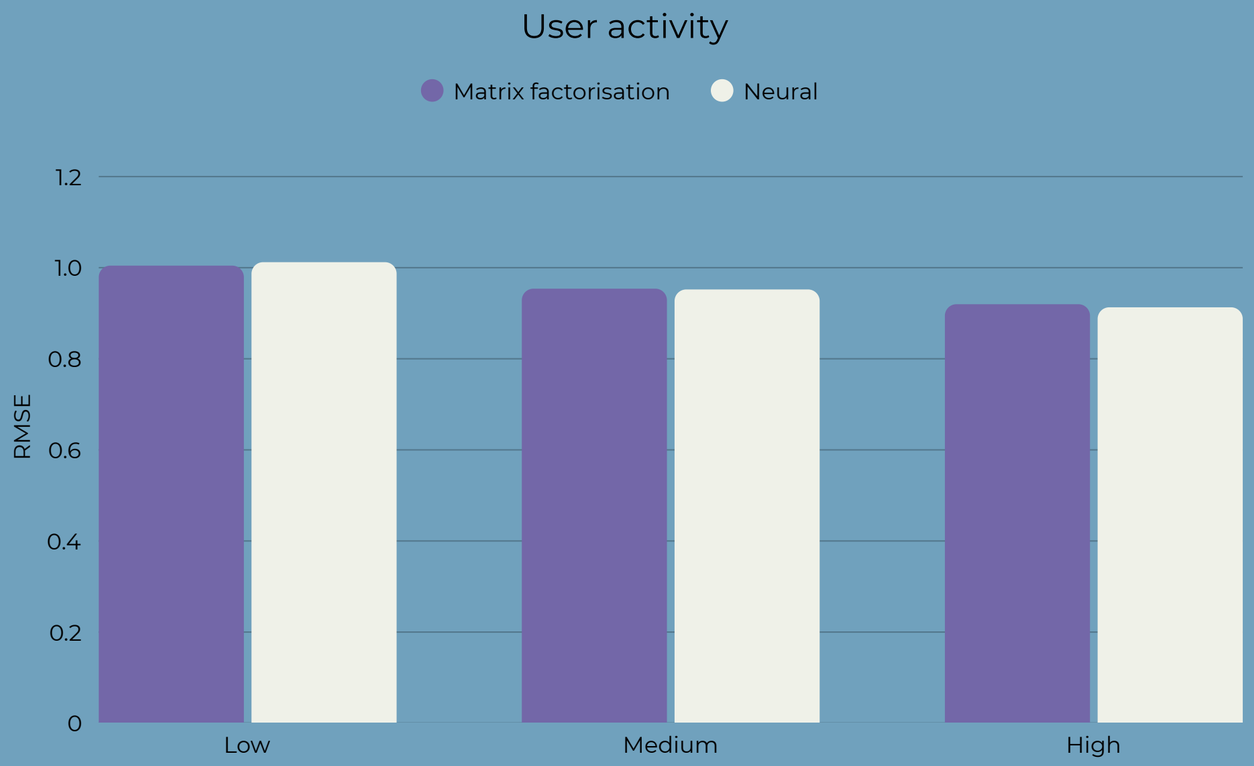
Rating accuracy and ranking results

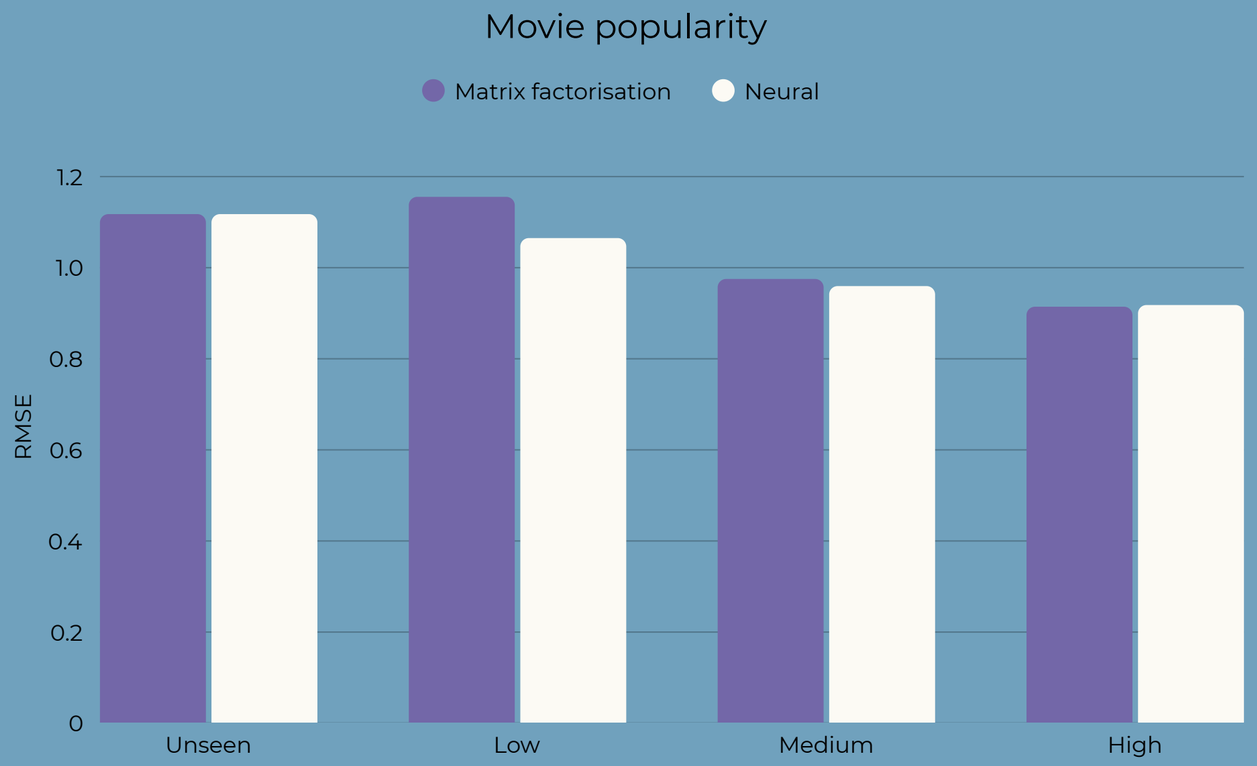
Model	RMSE	MAE	NDCG@10
Global mean	1.1257	0.9448	0.6198
User mean	1.0438	0.8364	0.6198
Movie mean	1.0271	0.8190	0.7994
Matrix factorisation	0.9372	0.7391	0.8103
Neural recommender	0.9332	0.7330	0.8053

Neural model: slightly lower rating error
Matrix factorisation: slightly higher NDCG

Five-fold test averages. Statistical significance remains untested.

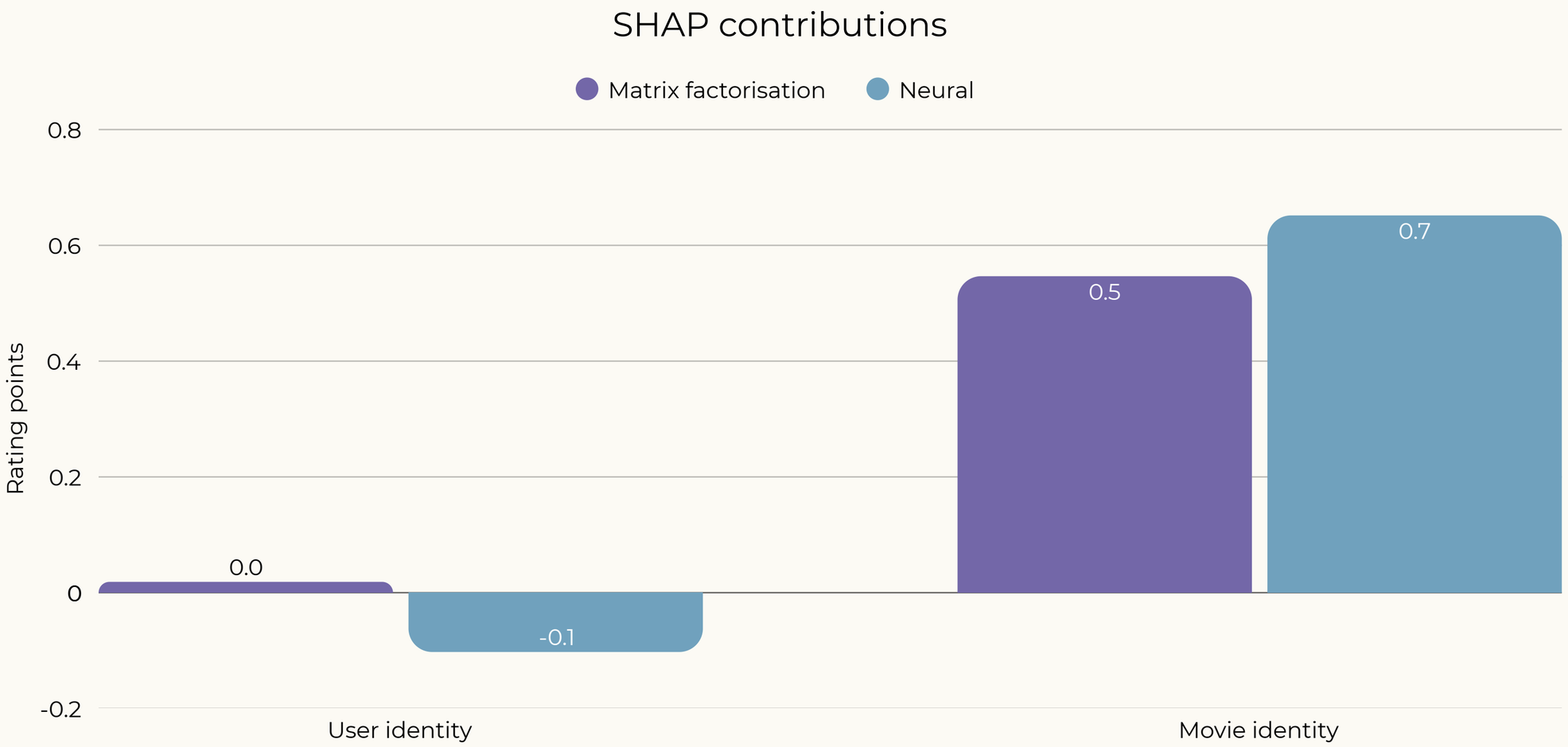
Errors by user activity and movie popularity





Less-popular movies: 1,928 test ratings. Unseen movies: 236. Both models use the same fallback.

Local explanations



User 608
Movie 190

Actual rating: 4
MF: 4.1747
Neural: 4.1378

Background predictions: MF 3.6093, neural 3.5885. SHAP does not identify the causes of preference.

Recommendation demonstration

Rank	Matrix factorisation	Neural recommender
1	Trainspotting	Schindler’s List
2	Brazil	A Close Shave
3	Mighty Aphrodite	The Wrong Trousers
4	Rosencrantz and Guildenstern Are Dead	Lawrence of Arabia
5	The Apartment	The Usual Suspects

Notebook: show_recommendations(user_id=608)

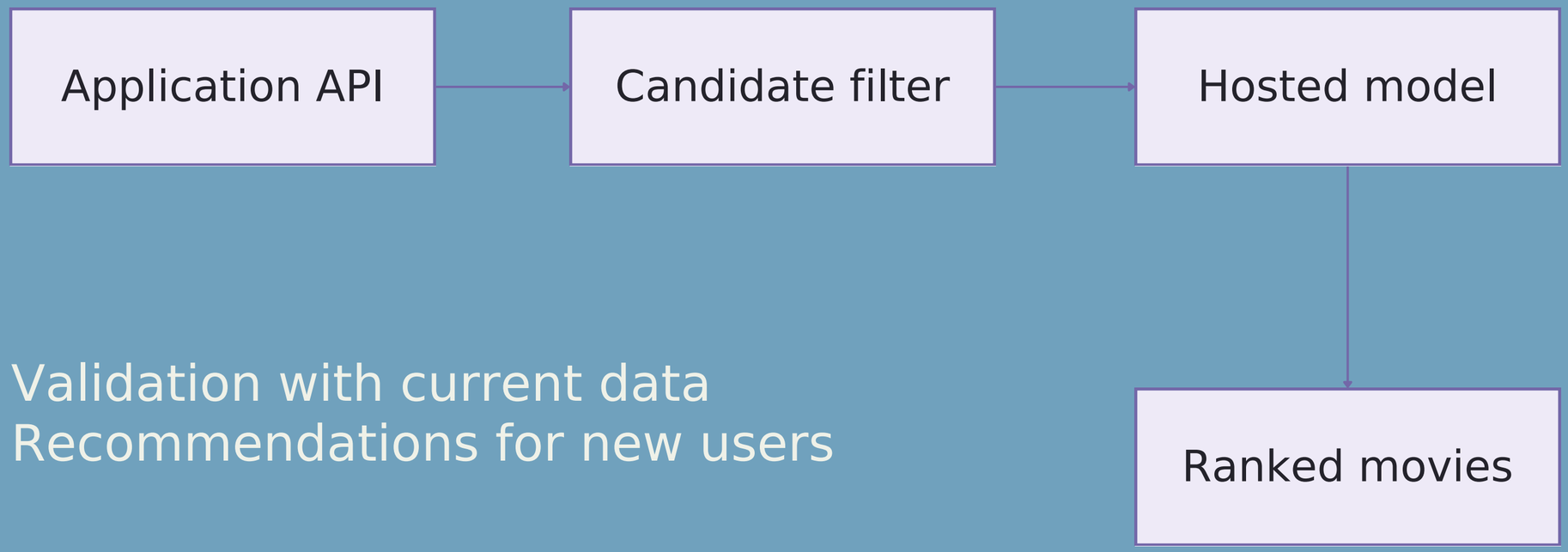
The demo scores a broader set of movies. NDCG uses only movies with known test ratings.

Ethical implications

Risk	Consequence	Proposed safeguard
Rating histories that could identify users	Privacy loss	Minimise data and restrict access
Sparse evidence	Uneven prediction errors	Compare errors by available training evidence
False positive	Promotes a disliked movie	Feedback and dismissal controls
False negative	Limits discovery	Evaluate exploration and exposure

Consent depends on the proposed use. GroupLens requires permission for commercial use and redistribution.

Deployment proposal



Validation with current data
Recommendations for new users

Monitor errors and recommendation exposure
Review model updates and retain a rollback option

Deployment is a proposal. Response time, capacity, cost and new-user quality require testing.

Model recommendation

Matrix factorisation for an initial pilot

Slightly higher NDCG in this evaluation

Average selected trial time: MF 1.167 s, neural 6.354 s

Neural model: lower error for less-popular movies

Before deployment, test future ratings and recommendations for new users.

References

Harper and Konstan (2015)
The MovieLens Datasets: History and Context

Koren, Bell and Volinsky (2009)
Matrix Factorization Techniques for Recommender Systems

He et al. (2017)
Neural Collaborative Filtering

Lundberg and Lee (2017)
A Unified Approach to Interpreting Model Predictions

GroupLens, PyTorch, scikit-learn, SHAP and AWS documentation